SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1520

COURSE OUTCOME COVERED
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct
I A. Tharun Kumar Reddy (192372068) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A. Tharun

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

Application Based Question.

(1) Cloud Auto-Scaling and Load Balancing:-

* Handling Demand Efficiently:-

Load Balancers act as traffic cops, distribution incoming user requirt, evenly across Multiple servers so No single server Get overhelmed.

Auto-Scaling dynamically adjusts the number of active based on real-time traffic demand.

* Reducing Downtime & Improving users Experience:-

By automatically scaling out during a surge, the system prevents servers from crashing due to resources Exhaustion. Minimizing downtime, for users, this means consistent Page.

* Role of Monitoring Tools:-

Tools like Aws cloudwatch (or) Datadog continously track Metrics like Cpu utilization, Memory, usage and Request counts, when a Metric crosses a Predefined threshold, it trigeers the auto-the Scaling Policy to add (or) remove to the speed instances.

(2) Cloud storage and Disaster Recovery:-

* Durability and Data Redundancy:-

Cloud Providers achieve high durability often a 99.999999% or "11 Nines") by automatically replicating data across Multiple Geographically separated datacentre Availability zones.

* Protecting Sensitive Backup Data:-

- Encryption
- Access control.

* Cost Advantages:-

Cloud storages operates on a Pay-as-you-go Model eliminating Massive upfront Capital Expenditure (CapEX) for buying physical Hard drives and Servers.

* Disaster Recovery use case:-

- A ransomware attack encrypts an organizations on-Premises databases, using a cloud diaster recovery setup, the organization can instantly restore.

(5) (3)

Content Delivery Network (CDNs) and Global Availability:-

- Latency Reduction:-

Latency is the delay caused by data traveling long physical distance. CDNs solve this by caching static website assets (images, videos, code files).

- Bandwith optimization:-

Delivering content locally reduces the load on the company's origin server, leading to drastically faster page loading speeds.

Enhancing users experience via Edge Servers:-

- Edge Routing:-

Edge servers specialized proxy servers deployed globally in points of presence (PoPs).

- Localized Processing:-

When a user request content, the edge of a server closest to them fulfills it immediately. This ensures that users London.

Traffic Managment:-

Anycast routing and Golbal load balancers the Automatically detect.

(5)

(4) Regulatory Compiliance, Auditing, and Data Sovereignty:-

Supporting compliance and Auditing:-

- Certified Infrastructure:-

Cloud Providers Maintain extensive Compilance the Certifications such Soc 2, ISO, 27001, HIPAA, and PCI-DSS.

- Shared Responsibility:-

The Provider ensures the underlying cloud infrastructure is compliant, saving companies from the Compiliant.

- The Security Triad:-
  - Logging
  - Monitoring
  - Encryption.

- Local Data Residency:-

Data Sovereignty laws (like the EU's GDPR) Require data to remain within specific National borders companies ensures compliance by explicitly storage.

(5)

Containerization orchestration, and Microservices:-

Containers Vs. Virtual Machines (VMs)

- **Resource Efficiency:-**

Virtual Machines requires a full Guest operating System for every instance, making them heavy and resources, Containers share the host Machine's OS kernel, making them lightweight.

- **Performance:-**

Containers boot up in seconds and consume a fraction of the RAM and CPU compared to VMs, allowing much higher application density on the a same physical Server.

- **Simplifying Development and Scaling via Orchestration:-**

  - Automated Management
  - Self-Healing and Scaling.
  - Environment
  - Multi-cloud flexibility

Example, Scenario, Microservice setup, flight search Services.